

Experiment (6)

Telnet

6.1 Objectives

To achieve Telnet protocol and understanding mechanism of it's work

6.2 Equipment / Program Requirements:

Ensp

Cisco Packet Tracker

6.3 Experiment Setup:

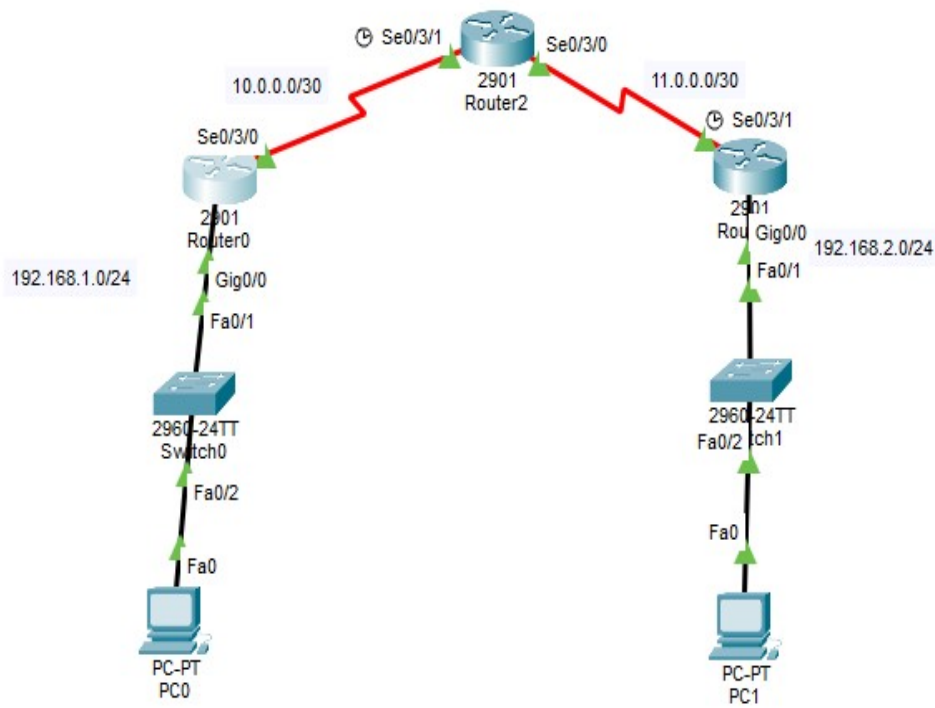
First make topology as shown under

Second divide it to 3 areas different and achieve ospf protocol and make configuration of ips in each router

Third enable telnet protocol with username and password

6.4 Simulation Results:

Topology:



Router0 Configuration:

Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router(config)#ip add
Router(config)#int g0/0
Router(config-if)#ip add
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int se0/3/0
Router(config-if)#ip add
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.1.0 0.0.0.255 area1
^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.1.0 0.0.0.255 area 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#
00:23:39: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/3/0 from LOADING to FULL,
Loading Done

Router(config-router)#
Router(config-router)#do show ip route
Codes: L - local, C - connected, S - static, D - DED, M - mobile, R - RCD
```

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Router0 Ospf database:

Router0

Physical
Config
Attributes

IOS Command Line Interface

```

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 192.168.1.0 0.0.0.255 area1
^
% Invalid input detected at '^' marker.

Router(config-router)#network 192.168.1.0 0.0.0.255 area 1
Router(config-router)#network 10.0.0.0 0.0.0.3 area 0
Router(config-router)#
00:23:39: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/3/0 from LOADING to FULL, Loading Done

Router(config-router)#
Router(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
    11.0.0.0/30 is subnetted, 1 subnets
O       11.0.0.0/30 [110/128] via 10.0.0.2, 00:04:44, Serial0/3/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O IA    192.168.2.0/24 [110/129] via 10.0.0.2, 00:03:37, Serial0/3/0

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Router1 configuration:

Router1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se0/3/1
Router(config-if)#ip add
Router(config-if)#ip address 11.0.0.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

Router(config-if)#int g0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config-if)#ip add
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shh
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#router ospf 1
Router(config-router)#router-id 3.3.3.3
Router(config-router)#network 192.168.2.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 192.168.2.0 0.0.0.255 area 2
Router(config-router)#network 11.0.0.0 0.0.0.255 area 0
Router(config-router)#
00:24:38: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/3/1 from LOADING to FULL,
Loading Done

Router(config-router)#do show ip ospf neighbor
```

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Router1 routing table:

Physical Config **CLI** Attributes

IOS Command Line Interface

```

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 1 subnets
O   10.0.0.0/30 [110/128] via 11.0.0.1, 00:00:23, Serial0/3/1
  11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   11.0.0.0/30 is directly connected, Serial0/3/1
L   11.0.0.2/32 is directly connected, Serial0/3/1
O IA 192.168.1.0/24 [110/129] via 11.0.0.1, 00:00:23, Serial0/3/1
   192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.2.0/24 is directly connected, GigabitEthernet0/0
L   192.168.2.1/32 is directly connected, GigabitEthernet0/0

Router(config-router)#do show ip ospf database
      OSPF Router with ID (3.3.3.3) (Process ID 1)

      Router Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum Link count
1.1.1.1        1.1.1.1       115         0x80000002   0x002c76 2
3.3.3.3        3.3.3.3       48          0x80000002   0x00bfcf 2
2.2.2.2        2.2.2.2       48          0x80000004   0x004e8f 4

      Summary Net Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum
192.168.1.0    1.1.1.1       240         0x80000001   0x00b936
192.168.2.0    3.3.3.3       44          0x80000001   0x007175

      Router Link States (Area 2)

Link ID        ADV Router    Age         Seq#          Checksum Link count
3.3.3.3        3.3.3.3       48          0x80000002   0x00b31b 1

      Summary Net Link States (Area 2)

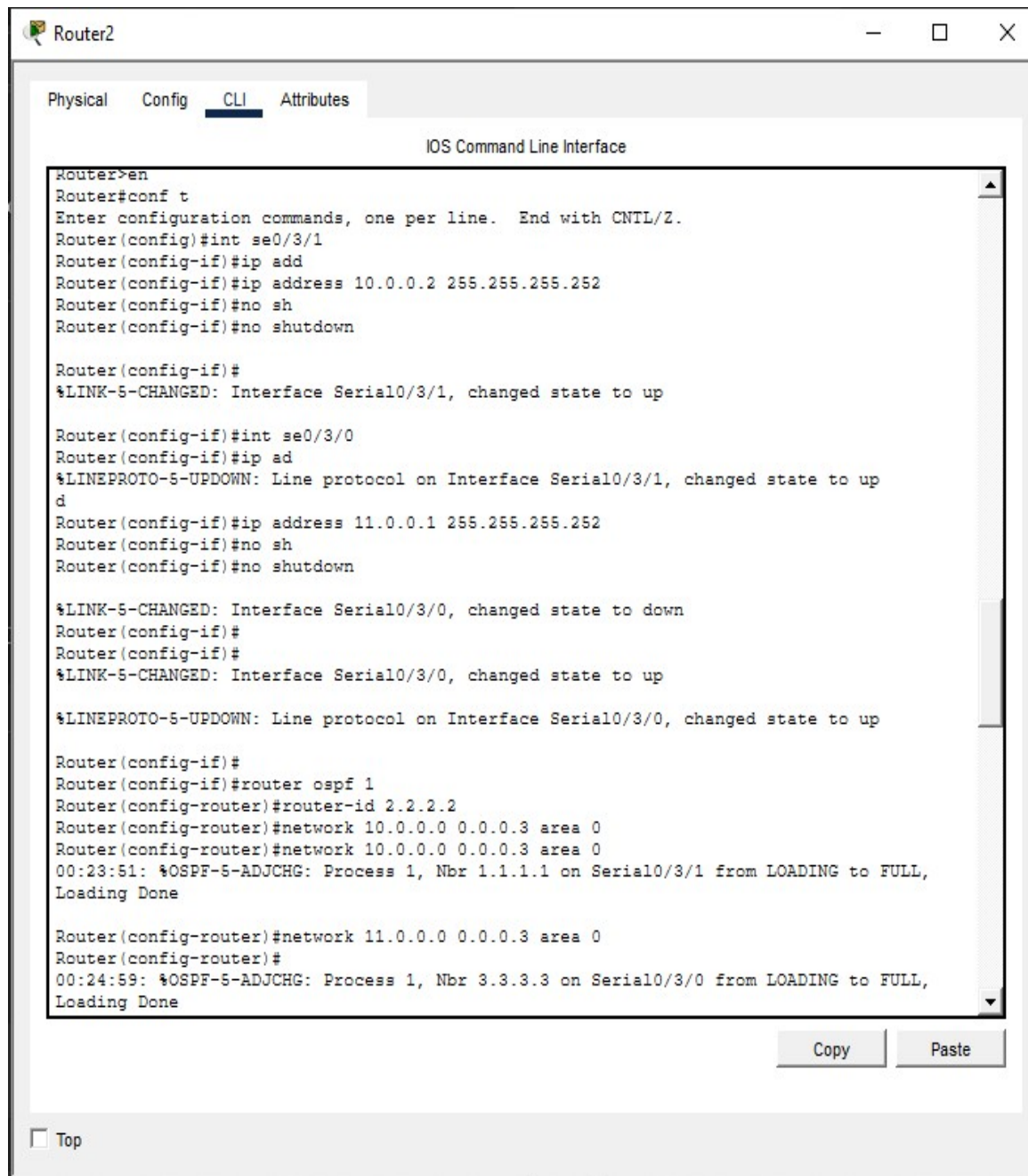
Link ID        ADV Router    Age         Seq#          Checksum
11.0.0.0       3.3.3.3       39          0x80000001   0x0010fa
10.0.0.0       3.3.3.3       39          0x80000002   0x009d2d
192.168.1.0    3.3.3.3       39          0x80000003   0x007de7
Router(config-router)#

```

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☐ Top**Router2 configuration:**



Router2 ospf database:

Router2

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

E1 - OSPF external type 1, E2 - OSPF external type 2, E - AS
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/1
L       10.0.0.2/32 is directly connected, Serial0/3/1
  11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/3/0
L       11.0.0.1/32 is directly connected, Serial0/3/0
O IA 192.168.1.0/24 [110/65] via 10.0.0.1, 00:04:19, Serial0/3/1
O IA 192.168.2.0/24 [110/65] via 11.0.0.2, 00:03:07, Serial0/3/0

Router(config-router)#do show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address         Interface
1.1.1.1        0     FULL/ -         00:00:37    10.0.0.1        Serial0/3/1
3.3.3.3        0     FULL/ -         00:00:34    11.0.0.2        Serial0/3/0
Router(config-router)#do show ip ospf database
          OSPF Router with ID (2.2.2.2) (Process ID 1)

          Router Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum Link count
1.1.1.1        1.1.1.1       277         0x80000002    0x002c76 2
2.2.2.2        2.2.2.2       210         0x80000004    0x004e8f 4
3.3.3.3        3.3.3.3       210         0x80000002    0x00bfcf 2

          Summary Net Link States (Area 0)

Link ID        ADV Router    Age         Seq#          Checksum
192.168.1.0    1.1.1.1       402         0x80000001    0x00b936
192.168.2.0    3.3.3.3       205         0x80000001    0x007175
Router(config-router)#

```

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Router2 routing table:

```
Router(config-router)#
00:24:59: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/3/0 from LOADING to FULL,
Loading Done

Router(config-router)#do show ip router
show ip router
^
% Invalid input detected at '^' marker.

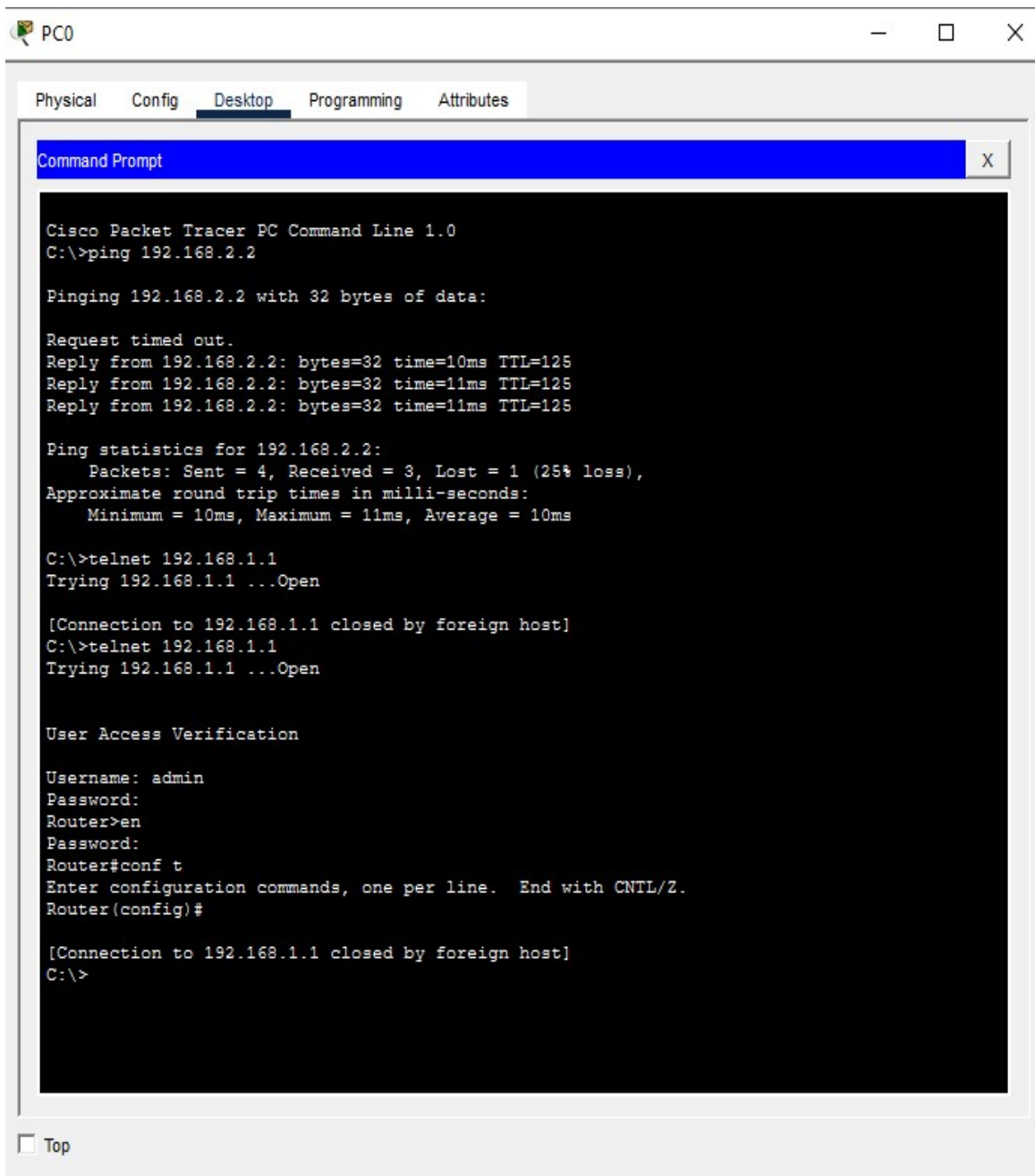
Router(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/1
L       10.0.0.2/32 is directly connected, Serial0/3/1
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/3/0
L       11.0.0.1/32 is directly connected, Serial0/3/0
O IA 192.168.1.0/24 [110/65] via 10.0.0.1, 00:04:19, Serial0/3/1
O IA 192.168.2.0/24 [110/65] via 11.0.0.2, 00:03:07, Serial0/3/0

Router(config-router)#do show ip ospf neighbor
```

Telnet & Testing with Ping Protocol:



6.5 Conclusions:

It's Important to use Telnet Protocol to make remote access to other devices